

Information Security

Module 01

Duration: 26:44

Video: <https://www.youtube.com/watch?v=q4BLwk2kOGs>

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my dear students hello I welcome you all in our series of lectures on cyber security today we are going to discuss about social and ethical issues of information security so let's start with it ethics is a field of study that is concerned with distinguishing right from wrong and good from bad it analyzes the morality of human behaviors policies laws and social structures ethicists attempt to justify their moral judgments by reference to ethical principles of theories that attempt to capture our moral intuitions about what is right and wrong the two theoretical approaches that are most common in ethics are consequentialism and analogy consequentialist approaches assume that actions are wrong to the

extent that they have bad consequences whereas deontological approaches assume that people have moral duties that exist independently of any good or bad consequences that their actions may have ethical principles often inform legislation but it is recognized in ethics that legislation cannot function as a substitute for Morality it is for this reason that individuals and corporations are always required to consider not only the legality but also the morality of their actions ethical analysis of security and privacy issues in Information Technology primarily takes place in computer ethics which emerged in the 1980s as a field computer ethics analyzes moral responsibilities of computer professionals and computer users and ethical issues in public policy for information technology development and

use it asks such questions as is it wrong for corporations to read their employees email or is it morally permissible for computer users to copy copyrighted software or should people be free to put controversial or pornographic content online without censorship ethical issues and questions like these require moral or ethical analysis analysis in which the moral dilemmas contained in these issues are clarified and solutions are proposed for them moral analysis aims to get clear on the facts and values in such cases and to find a balance between the various values rights and interests that are at stake and to propose or evaluate policies and courses of action now the moral importance of computer

security computer security is a field of computer science concerned with the application of security features to computer systems to provide protection against the unauthorized disclosure manipulation or deletion of information and against denial of service the condition

resulting from these efforts is also called computer security the aim of computer Security Professionals is to attain protection of valuable information and system resources a distinction can be made between the security of the system resources and the security of information or data the first may be called system security and the second information security or data security system security is the protection of the hardware and software of a computer system against malicious

programs that sabotage system resources information security is the protection of data that resides on disk drives on computer systems or is transmitted between systems information security is customarily defined as concerned with the protection of three aspects of data confidentiality integrity and availability now how does computer security pose ethical issues as explained earlier ethics is mostly concerned with rights harms and interest we may therefore answer this question by exploring the relation between computer security and rights harms and interest now the important questions are what morally important benefits can computer security bring what morally important harms or violations of moral rights can result from a lack of computer security or can computer

security also cause harms or violate rights instead of preventing and protecting them a first and perhaps most obvious harm that can occur from breaches of computer security is economic harm when system security is undermined valuable hardware and software may be damaged or corrupted and service may become unavailable resulting in losses of time money and sources breaches of information security may come at an even higher economic cost valuable data may be lost or corrupted that is worth much more than the hardware on which it is stored and this may cause severe economic losses stored data may also have personal cultural or social value as opposed to the economic value that can be lost when data is corrupted or lost any type of loss of system or data security is more likely to cause some

amount of psychological or emotional harm breaches of computer security may even cause grave harms like injury and death this may occur in so-called safety critical systems which are computer systems with a component or realtime control that can have a direct life-threatening impact examples are computer systems in nuclear reactor control aircraft and air traffic control missile systems and medical treatment systems the corruption of certain other types of systems may also have life-threatening consequences in a more indirect way these may include systems that are used for design monitoring diagnosis or decision making for instance systems used for bridge design or medical diagnosis compromises of the confidentiality of information may cause

additional harms and rights violations third parties May compromise the confidentiality of information by accessing copying and disseminating it such actions May first of all violate property rights including intellectual property rights which are rights to own and use intellectual Creations such as artistic or literary works and Industrial designs the information may be exclusively owned by someone who has the right to determine who can access and use

the information and this right can be violated second compromises of confidentiality May violate privacy rights this occurs when information that is accessed includes information about persons that is considered to be private in addition to violations of property and privacy rights breaches of

confidentiality may also cause a variety of other harms resulting from the dissemination and use of confidential information for instance dissemination of internal members of a firm damages its reputation and compromises of the confidentiality of online credit card transactions undermines trust in the security of online Financial transactions and harms ebanking and Ecommerce activity compromises of the availability of information can when they are prolonged or intentional violate Freedom rights specifically rights to Freedom of Information and Free Speech Freedom of Information is the right to access and use public information shutting down Vital Information Services could violate this right to information in addition computer

networks have become important as a medium for speech websites email bulletin boards and other services are widely used to spread messages and communicate with others when access to such Services is blocked for instance through denial of service attacks or hijackings of websites such acts are properly classified as violations of free speech computer security measures normally prevent harms and protect rights but they can also cause harm and violate rights notably security measures may be so protective of information and system resources that they discourage or prevent stakeholders from accessing information or using services security measures may also be discriminatory they may wrongly exclude

certain classes of users from using a system or may wrongly privilege certain classes of users over others next is ethical issues in computer security number one hacking and computer crime a large part of computer security is concerned with the protection of Computer Resources and data against unauthorized intention breakings or disruptions such action is often called hacking hacking is the use of computer skills to gain unauthorized access to Computer Resources hackers are highly skilled computer users that use their talents to gain such access and often from communities or networks with other hackers to share knowledge and data hacking is often also defined more negatively as the gaining of such

unauthorized access for Malaysia purposes to steal information and software or to corrupt data or disrupt system operations self-identified hackers however make a distinction between non-malicious break-ins which they describe as hacking and malicious and disruptive break-ins which they call cracking self-identified hackers often justify their hacking activities by arguing that they cause no real harm and instead have a positive EMP fact the positive impact of hacking they argue is that it frees data to the benefit of all and improve systems and software by exposing security holes these considerations are part of what has been called The Hacker ethic or hacker code of ethics which is a set of principles that guide the activity of many

hackers such principles include convictions that information should be free that access to computers should be unlimited and total and that activities in cyberspace cannot do harm in the real world the belief that information should be free runs counter to the very notion of intellectual property and would imply that creators of information would have no right to keep it to themselves and have no opportunity to make a profit from it it would more however fundamentally undermine priv privacy and would undermine the integrity and accuracy of information as information could be modified and changed at will by anyone who would access it there are many varieties of computer crime and not all of them compromise computer security there are two major types of

cyber crime that compromise computer security cyber trespass which is defined as the use of Information Technology to gain unauthorized access to computer systems or password protected websites and cyber vandalism which is the use of Information Technology to unleash programs that disrupt the operations of computer networks or corrupt data number two cyber terrorism and information Warfare a recent concern in computer and National Security has been the possibility of cyber terrorism which is defined as the execution of politically motivated hacking operations intended to cause grave harm that is resulting in either loss of life or severe economic loss or both the possibility of major attacks on information infrastructure intending to

debilitate or compromise this infrastructure and harm economic industrial or social structures dependent on it has become a major concern since the 9/11 tax such attacks could be both foreign and domestic cont y exist on the proper scope of cyber terrorism where should the boundaries be drawn between cyber terrorism cyber crime and cyber vandalism should a teenager who releases a dangerous virus that turns out to cause major harm to government computers be persecuted as a cyber terrorist or are politically motivated hijackings of the homepages of major organizations acts of cyber terrorism a distinction between cyber terrorism and other kinds of cyber attacks may be found in its political nature cyber

terrorism consists of politically motivated operations that aim to cause harm there is a distinction between cyber terrorism and hactivism activism are hacking operations against an Internet site or server with the intent to disrupt normal operations but without the intent to cause serious damage activists may make use of email bombs low grade viruses and temporary homepage hijackings there are politically motivated hackers who engage in a form of electronic political activism that should be distinguished from terrorism information Warfare is an extension of ordinary Warfare in which combatants use information and attacks on information and information systems as tools of warfare information Warfare May include the use of information media

15 minutes, 1 second to spread propaganda the disruption jamming or hijacking of communication infrastructure or propaganda fears of the enemy and hacking into computer systems that control vital infrastructure for example oil and gas pipelines electric power grids

or rail infrastructure now what are the moral responsibilities of Information Security Professionals Information Security Professionals are individuals whose job is to maintain system and information security by standing of their profession they have a professional responsibility to assure the correctness reliability availability Safety and Security of all aspects of information and information systems the discussion makes clear that this responsibility has a moral Dimension professional activities in

computer security may protect people from AI important harms but could also cause such harms and may either protect or violate people's moral rights in case of safety critical systems the decisions of important Security Professionals may even be a matter of life or death that information Security Professionals have moral responsibilities as part of their profession is reflected in CES of Ethics used by various organizations for computer and information security these CES of Ethics rarely go into detail however on the moral responsibilities of Information Security Professionals in specific situations for instance the code of ethics of Information Systems security Association an international organization of Information Security

Professionals and practitioners only states that members should perform all professional activities and duties in accordance with all applicable laws and the highest ethical principles but does not go on to specify what these ethical principles are or how they should be applied and balanced against each other in specific situations for Information Security Professionals as well as for other computer professionals who have a responsibility for computer security a code of ethics clearly is not enough to appreciate the moral dimension of their work and to cope with moral dilemmas in it they require training in information security ethics such training help professionals to get clear about interests rights and

moral values that are at stake in computer security to recognize ethical questions and dilemmas in their work and to balance different moral principles in resolving such ethical issues number three information privacy and ethics what is privacy and why is it important in Western societies a broad recognition exist of a right to personal privacy the right to privacy was first defended by the American Justice Samuel Brandeis who defined privacy as the right to be let alone privacy is a notion that is difficult to Define and many more precise definitions have since been presented often the right to privacy is defined as a right of individuals to Control Access or interference by others into their Private Affairs privacy is held to be valuable

for several reasons most often it is held to be important because it is believed to protect individuals from all kinds of external threats such as defamation ridicule harassment manipulation blackmail theft subordination and exclusion it has also been argued that privacy is a necessary condition for autonomy without privacy people could not experiment in life and develop their own personality and own thoughts because they would constantly be subjected to the Judgment of others the right to privacy has also been claimed to protect other rights such as abortion rights and the right to sexual expression privacy moreover has been claimed to

have social value in addition to individual value it has for instance been held to be essential for for maintaining democracy the right to privacy is not

normally held to be absolute it must be balanced against other rights and interest such as the maintenance of public order and national security privacy rights may also vary in different contexts there is for example a lesser expectation of privacy in the workplace or in the public sphere than there is at home an important principle used in privacy protection in Western Nations is that of informed consent it is often held that citizens should be informed about how organizations plan to store use or exchange their personal data and that they should be asked for their consent people can then voluntarily give up their privacy if they choose number four information technology and privacy privacy is a value in modern

societies that corresponds with the ideal of the autonomous individual who is free to act and decide his own destiny yet modern societies are also characterized by surveillance a practice that tends to undermine privacy surveillance is a systematic observation of groups of people for specific purposes usually with the aim of exerting some form of influence over them the state engages in surveillance to protect National Security and to fight crime and the modern Corporation engages in surveillance in the workplace to retain control over the workforce computerization from the 1960s onward has intensified surveillance by increasing its scale ease and speed surveillance is partially delegated to computers that help in collecting processing and exchanging

data computers have not only changed the scale and speed of surveillance they have also made a new kind of surveillance possible data wance which is the large scale computerized collection and processing of personal data in order to monitor people's action and Communications more and more information technology is not just used to record and process static information about individuals but to record and process their actions and communication new detection Technologies like smart Clos circuit televisions Biometrics and intelligent user interfaces and new data processing techniques like data mining further exaggerate this trend the ease with which surveillance now takes place has made it a generalized activity that is routinely performed in all kinds of settings by

different kinds of organizations corporations for instance have extended surveillance from their workplace to their customers called consumer surveillance in addition the 911 terrorist attacks have drastically Ally expanded surveillance activities by the state many privacy disputes in today's society result from tensions between people's right to privacy and state and corporate interest in surveillance in the Information Society privacy protection is realized through all kinds of information privacy laws policies and directives or data protection policies these policies regulate the harvesting processing usage storage and exchange of personal data they're often overtaken however by new developments in

technology however privacy protection has also become a concern in the design and development of Information Technology number five privacy in public it is sometimes believed that privacy is a right that people have when they are in private places like homes private clubs and restrooms but that is minimized or forfeited as soon as they enter public space when you walk in public streets or are on the road with your car it is sometimes believed you may retain the right not to be seized and searched without probable cause but your appearance and behavior may be freely observed surveilled and registered many privacy Scholars however have argued that this position is not fully tenable and that people have privacy rights in public areas that are incompatible with certain registration and surveillance

24 minutes, 1 second practices the problem of privacy in public applies to the tracking recording and surveillance of public appearances movements and behaviors by individuals and their vehicles techniques that are used for this including video surveillance including smart CCTV for facial recognition infrared cameras satellite surveillance GPS tracking are F ID tagging electronic checkpoints mobile phone tracking audio bugging and mate intelligence techniques number six the last but not the least is biometric identification Biometrics is the identification or verification of someone's else identity on the basis of psychological or behavioral characteristics biometric Technologies provide a reliable method of access control and personal identification for

governments and organizations however Biometrics has also raised privacy concern widespread use of Biometrics would have the undesirable effect of eliminating anonymity and pseudonymity in most daily transactions because people would leave unique traces everywhere they go more however the biometric monitoring of movements and actions gives the monitoring organization insight into a person's behaviors which may be used against that person's interests in addition many people find biometrics distasteful because it involves the recording of unique and intimate aspects of a person and because biometric identification procedures are sometimes invasive of bodily privacy the challenge for Biometrics is therefore to develop techniques and policies that optimally protective of

personal privacy with this our today's lecture is over concluding the lecture we have learned about what are ethics importance of computer security and various social and ethical issues hope today's lecture is clear to you in the next lecture we shall learn about Information Security Management till then goodbye have a nice day